Published: 03 September 2025

Citation in BibTeX format

GoodIT '25: International Conference on Information Technology for Social Good
September 3 - 5, 2025
Antwerp, Belgium

Conference Sponsors:
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RESEARCH-ARTICLE

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GoodIT '25: Proceedings of the 2025 International Conference on Information Technology for Social Good (September 2025)
https://doi.org/10.1145/3748699.3749804
ISBN: 9798400720895

GeoMedia: A Mobile Platform for Location-Aware and Time-Sensitive Social Content

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Abstract

GeoMedia is an open-source mobile platform designed for sharing multimedia content—such as text, images, audio, and video—anchored to real-world geographic locations. By leveraging GPS data and a map-centric user interface, GeoMedia offers a seamless integration between digital content and physical space, enabling users to publish and explore posts directly on an interactive map. Additionally, GeoMedia enables users to view and post exclusive location-based media, which can only be accessed within a specific time frame or geographical range, reinforcing the connection between digital experiences and real-world presence. This paper presents the motivation behind GeoMedia, details its system architecture and implementation, and explores potential use cases and future development trajectories.

CCS Concepts

• **Information systems** → **Location based services**; • **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; • **Networks** → *Location based services.*

Keywords

Location-based social networks, Mobile app development, Geotagging, Multimedia sharing, Interactive maps

ACM Reference Format:
Alberto Morini, Lorenzo Perinello, and Claudio E. Palazzi. 2025. GeoMedia: A Mobile Platform for Location-Aware and Time-Sensitive Social Content. In *International Conference on Information Technology for Social Good (GoodIT '25), September 03–05, 2025, Antwerp, Belgium.* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3748699.3749804

1 Introduction

Over the past few decades, mobile applications have experienced explosive growth, increasingly replacing traditional desktop software and web browsing. This transformation is not only technological but also cultural, reflecting the shift toward a digital society [22]—one in which everyday activities, social interactions, and information flows are deeply embedded in digital infrastructure and mobile technologies. This transformation is driven by several key factors: the widespread adoption of mobile devices [15, 31], their growing computational power [13], increased battery life [8], and an expansion of use cases [19]. Modern smartphones offer a wider range of functionalities compared to conventional computers, largely due to their built-in sensors—such as gyroscopes, GPS modules, and light sensors—and actuators like vibration motors and cameras [33]. These features have enabled a richer, more interactive user experience and opened the door to innovative applications in diverse domains, including health [4], education [21], and serious games [23]. Applications that interact with the physical world—through sensors or cameras—and mobile games have become particularly prevalent [27]. Prominent examples such as Google Maps, Waze, Strava, Snapchat, and Instagram illustrate the popularity and ubiquity of such mobile apps, each boasting millions of downloads.

The shift from browser-based to mobile-based internet use has been striking. In 2011, only 6% of users accessed the internet via mobile phones; today, that figure exceeds 93%, representing approximately 4 billion people [11]. This trend is mirrored in social media usage: over 80% of Facebook users now access the platform exclusively through mobile devices, a dramatic increase from 30% in 2014 [20, 30]. As a result, many companies are prioritizing mobile platforms to deliver personalized, engaging experiences. To achieve this, they increasingly rely on sensor data to infer user behavior and preferences. For instance, GPS data can reveal movement patterns [29], while gyroscopes can detect specific gestures or actions [7]. However, this data collection raises several challenges, including faster battery depletion [14], higher data consumption [25], and significant privacy concerns [10].

As mobile hardware continues to evolve, there is an increasing convergence between physical and digital spaces. Location-aware technologies now support a wide spectrum of applications, from real-time navigation and location-based recommendations to augmented reality games and immersive media experiences. This shift has transformed how individuals interact not only with their devices but also with their surroundings, turning smartphones into context-sensitive tools capable of enhancing real-world interactions.

Despite this evolution, many existing location-based applications remain proprietary, rigid in functionality, or focused predominantly on data consumption rather than creation. There is a notable lack of open, extensible platforms that empower users to actively contribute multimedia content—such as audio, images, or video—anchored to specific physical locations. Furthermore, tools that enable users to collaboratively create and explore geo-tagged content in a transparent and privacy-conscious environment are scarce. These gaps present a clear opportunity for developing platforms that are both technically flexible and socially engaging.

Within this context, we introduce GeoMedia: an open-source mobile platform that enables users to share comments and multimedia content—such as photos, audio, or video—directly on a geographic

ACM ISBN 979-8-4007-2089-5/25/09
https://doi.org/10.1145/3748699.3749804

map, linking each post to the user's current location. GeoMedia aims to support interactive spatial storytelling, collaborative exploration, and personalized geolocation-based content sharing. Its modular architecture allows developers and researchers to extend the platform for various use cases, from tourism and urban art to education and field research.

The remainder of this paper is organized as follows: Section 2 reviews relevant related work in the domain of location-based social networks and geolocation in mobile applications. Section 3 introduces the design and main features of the GeoMedia platform. Section 4 describes the system architecture, detailing the presentation, application, and data layers. Section 5 focuses on the development of the mobile application, including its main components, such as the map interface, exclusive posts, and media handling. Section 6 presents the backend server implementation and its integration with the database, while Section 7 outlines the database schema and the use of stored procedures to manage data efficiently. Finally, Sections 8 and 9 present directions for future work and conclude the paper, respectively.

2 Related Work

The integration of geolocation features into online social networks (OSNs) has been widely explored in both commercial platforms and academic literature. In this section, we categorize related efforts into three main domains: commercial applications, academic research on geospatial interaction, and privacy and ethical considerations. We then discuss how GeoMedia distinguishes itself from existing solutions.

Location-Based Social Applications

Several mainstream applications combine social networking with geospatial functionalities. Google Maps exemplifies this integration by offering navigation, real-time traffic data, and immersive visualizations through Street View. The Local Guides program [12] incorporates crowdsourcing dynamics—encouraging users to submit reviews and photos—thereby blending location-based services with OSN-like user engagement. Another popular platform was Foursquare City Guide, which recommended nearby places based on user activity and preferences. Despite its discontinuation in 2024, its core features were preserved in Swarm, which continues to support spatial check-ins and location history tracking. Instagram [32] further popularized geotagging by allowing users to attach locations to posts. However, none of these platforms offer real-time filtering of content based on the physical location of the viewer, nor do they support features such as time-limited or geo-fenced visibility, which GeoMedia introduces as foundational capabilities.

Academic landscape

In the academic domain, researchers have studied how geolocation data can enable new forms of urban understanding and social analysis. Roick and Heuser [28] proposed a comprehensive taxonomy for location-based social media data, categorizing services based on user intent, data structure, and spatial-temporal resolution. Their work provides a foundational framework for evaluating and comparing location-aware platforms, highlighting the analytical potential of geo-social data for urban studies and spatial behavior analysis. Cranshaw et al.'s Livehoods Project [9] leveraged 18 million Foursquare check-ins to identify socially meaningful regions within cities, reflecting how geographic data can be used to analyze local cultural dynamics. A similar work by Zheng et al. [38] explores how large-scale location-based social media data can be mined to understand urban dynamics, such as identifying popular regions and mobility patterns within cities. By leveraging GPS trajectories, their framework supports applications in urban planning, traffic management, and location recommendation. Zhang et al.[37] proposed a unified geo-social influence metric to evaluate user influence and discover influential events in location-based social networks (LBSNs). By integrating a penalized hitting time for social proximity and a power-law-based geographical weight function, they effectively captured the combined effects of social and spatial proximity. Their approximate algorithms, Global Iteration and Dynamic Neighborhood Expansion, demonstrated scalable performance on real-world LBSN data, enabling efficient identification of influential users and events. Another prominent research direction involves the use of gamification to enhance user engagement in geo-based platforms. For instance, Bujari et al. [5] introduce PhotoTrip, a system that leverages geo-tagged photos from Flickr and integrates gamification to identify and recommend lesser-known cultural heritage sites along travel routes. Unlike typical travel planners, PhotoTrip emphasizes user engagement through a gamified feedback mechanism to refine photo relevance and enhance recommendation quality. Prandi et al. [26] introduced Geo-Zombie, a gamified multimedia mobile application that leverages zombies and a geographical maps to foster civic engagement. Designed to crowdsource urban accessibility data, the app encourages users to report architectural barriers while engaging in a playful escape-from-zombies scenario. Their field trials with 50 participants showed that gamification, especially through intrinsic motivation like narrative immersion, significantly increased data contributions compared to non-gamified tools, suggesting that entertainment-driven apps can effectively support social inclusion initiatives. Such studies highlight the potential of location-aware applications to foster engagement, education, and collaborative knowledge creation.

Privacy, Ethics, and Design Challenges

The increasing ubiquity of geolocation raises significant privacy and ethical concerns. Tonetto et al. [35] reviewed over 100 studies on human mobility and found widespread inconsistency in the application of privacy principles, especially regarding data retention and user consent. The lack of transparency and control over location data remains a critical issue in many commercial applications. GeoMedia addresses this by incorporating visibility constraints directly into the content publishing workflow. Posts can be configured to expire after a set time or to be visible only within a defined radius, thereby promoting privacy-aware interactions. This user-centric control mechanism offers an alternative to the data-intensive practices seen in large-scale OSNs.

In summary, while existing platforms leverage location data for personalization or content discovery, they typically lack mechanisms for real-time, proximity-based interaction or spatial exclusivity. GeoMedia fills this gap by enabling location-bound and time-sensitive content sharing, fostering a deeper connection between digital media and the physical world. It is designed not only as a functional mobile platform but also as a research tool for experimenting with new paradigms of location-based communication and interaction.

3 GeoMedia

GeoMedia positions itself among existing OSNs, offering a way to share location-based content of various types—including text, images, and videos. The core of the app is a geographical map, where users can post multimedia content or discover posts shared by others. The mobile app, once properly allowed, automatically retrieves the user's location via the phone's GPS, seamlessly integrating the online social experience with the physical world. Furthermore, the app enables users to create exclusive content by restricting the visibility of a post to a specific time frame or a defined geographical area. GeoMedia thus provides a complete solution for sharing content tied to the user's real-world position, making the OSN inherently interactive and removing the need to manually select a location, as is often required in other social media platforms. The whole source code developed for this project is available on GitHub [24].

3.1 An example

An interesting scenario can be considered using the city of Padua to illustrate one of its well-known sayings: "città dei tre senza" (literally, "the city of the three without"). This expression refers to three notable things that Padua has—but in a metaphorical way:

- the lawn without grass: referring to Prato della Valle, which despite its name ("meadow of the valley"), features ornamental grass and is not a lawn in the traditional sense;
- the saint without a name: referring to Saint Anthony of Padua (Sant'Antonio da Padova), who was actually born in Portugal and only spent the last years of his life in Padua;
- the café without doors: referring to the historic Caffè Pedrocchi, which remained open day and night during wartime to offer shelter and respite during bombings.

In this context, the city of Padua could use GeoMedia to place digital markers at these locations, as shown in Figure 1, enabling the storytelling of each unique characteristic. At each marker, historical photographs could also be uploaded to show how these sites looked in the past, as illustrated in Figure 2.

3.2 Further use cases

Beyond historical storytelling, GeoMedia supports a variety of engaging and innovative applications that integrate digital content with real-world locations. Some notable use cases include the following, although not all have been fully implemented on the platform at this time:

- **Interactive experiences**: Users can follow guided narratives or quests that unfold as they move through specific geographic areas. These can be educational, cultural, or entertainment-oriented, offering dynamic, location-based content;
- **personalized dedications**: GeoMedia could support the creation of private, one-to-one messages or multimedia posts that are only visible to a specific recipient near a designated location. This allows for surprise greetings, commemorative messages, or intimate storytelling tied to meaningful places;
- **location-based games**: examples include digital scavenger hunts, where players uncover clues scattered across an area, and hide-and-seek-style games, in which users hide virtual

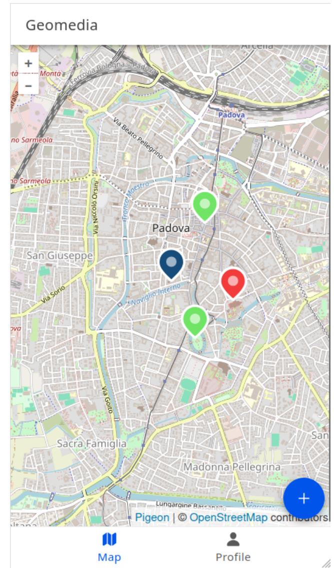

Figure 1: A screenshot of the GeoMedia map centered on the city center of Padua. It displays three different markers and the user's location (indicated by the blue marker).

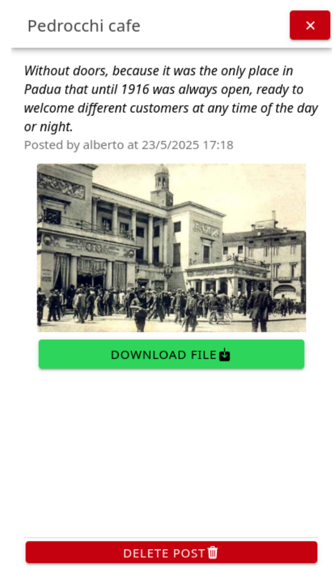

Figure 2: A screenshot showing the details of a post. Specifically, the post provides information about the Pedrocchi Café and includes an attached image that can also be downloaded.

content that others must physically approach to reveal. These features could foster exploration and social interaction;
- **virtual murals**: multiple users can contribute collaboratively to a digital artwork anchored to a real-world location. Through media uploads such as images, sketches, or videos, participants co-create evolving, site-specific art pieces, transforming urban or public spaces into shared interactive canvases.

Use Case	Number of Creators	Number of Viewers	Content Exclusivity	
			Geo-limited	Time-limited
Interactive experience	1+	Many	✓	×
Personalized dedication	1	1	✓	×
Location-based game	1+	Many	✓	✓
Virtual mural	Many	Many	✓	×

Table 1: Summary of GeoMedia use cases, participant scale, and exclusivity parameters. *"1"* refers to a single user, *"1+"* to one or more users, and *"Many"* to a broader group (numerosity greater than 1). A checkmark (✓) indicates the presence of a restriction (e.g. geo-limited or time-limited content), while a cross (×) indicates its absence.

4 Architecture

This section outlines the high-level, three-layer architecture of the GeoMedia system. As shown in Figure 3, it is composed of the presentation layer (the Android mobile app), the application layer (the backend), and the data layer (the database).

4.1 Presentation layer

The presentation layer is represented by the user interface of the mobile application. It allows users to view published comments or share their own content. In the latter case, the mobile app, as part of the presentation layer, is also responsible for retrieving data such as the current location, username, and other metadata.

4.2 Application Layer

In this project, the application layer is represented by an HTTP server, which handles incoming requests using appropriate functions and logic such as computing the distance from a given location to exclusive posts. The HTTP server exposes endpoints that the front-end (presentation layer) accesses by sending HTTP requests. For each endpoint, the server retrieve information by interrogate the Database Management System (DBMS) through the developed queries.

4.3 Data layer

The data layer consists of a DBMS, capable of performing CRUD operations (Create, Read, Update, Delete) on data stored in the database and structured in tables. In this project, the DBMS is also used to create SQL stored procedures [36], offering a more flexible way to modify `SELECT` or `INSERT` queries in future extensions without changing the code within the server program.

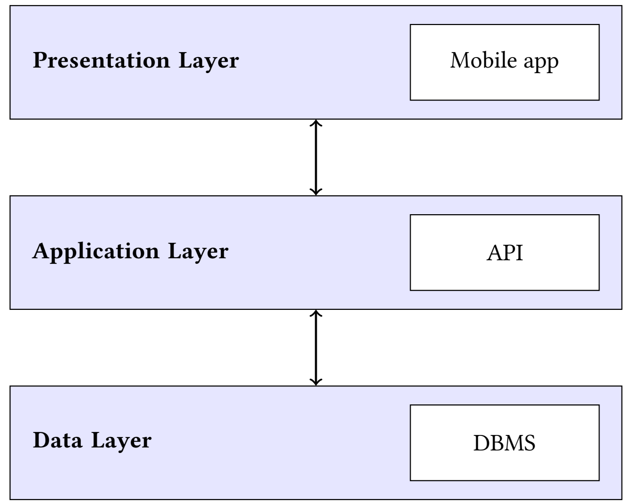

Figure 3: High-level three-layer architecture of GeoMedia, illustrating the separation of concerns between the Presentation Layer (mobile app interface), the Application Layer (backend API and logic), and the Data Layer (DBMS for data storage and retrieval).

5 Mobile Application

The Android mobile application is developed using the Ionic Framework [16], an open-source toolkit for building cross-platform mobile, web, and desktop applications using web technologies such as HTML, CSS, and JavaScript. One of the key advantages of the Ionic Framework is its ability to integrate with web frameworks and libraries, such as React, which was used in this project.

5.1 Key functionality

GeoMedia starts with a log-in component, where users can choose to create a new profile or to sign in whenever they already have an account. Once logged into the app, GeoMedia offers a bottom menu with the following sections:

- profile: in this tab, shown in Figure 4, users can view their posts as a list and open them individually. They can also modify backend server configuration parameters (e.g., IP address or HTTP port);
- map: in this section is the core of the application, offering navigation through a geographical map where users can explore comments and posts shared by others. Additionally, this tab provides access to the publish function, which allows users to create a new post based on their current coordinates. The post UI is shown in Figure 5.
  A single post is composed of the following attributes:
  - title;
  - comment;
  - multimedia content (optional);

– metadata automatically retrieved (e.g., longitude, latitude, username, date, etc.).

Figure 4: Screenshot of the GeoMedia profile page. The page provides access to settings and a logout option, along with a detailed list of the user's posts, including associated data and geographical coordinates. Each post also features a button that leads to a detailed view page.

5.2 Map component

This component is the core of the GeoMedia application. It is built upon Pigeon Map [2], which is a React library that encapsulates and provides a geographical interactive map; this library also allows to place markers in specific geographical location, which, in GeoMedia, represent a content posted by an user.

There are different types of location indicators:

- blue marker: indicates the current user's location, retrieved in real time via GPS;
- red marker: represents a post that contains only textual comments;
- yellow marker: represents a post that includes attached multimedia content;
- green marker: indicates an exclusive post, which is visible only at a specific date and time, or within a defined geographical range.

When the mobile application sends a request to the server to retrieve data, it provides the user's current location. The backend then filters only the relevant posts stored in the data layer and returns an HTTP response containing the corresponding content,

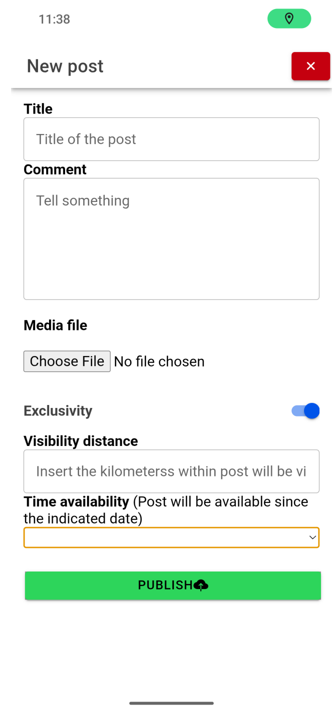

Figure 5: Screenshot of the post publication interface. It includes text input fields for the title and comment, an input for multimedia content, and sections for specifying geographical exclusivity (by setting the visibility radius in meters) and time availability (by setting a post expiration date).

along with their types and coordinates. The specific methods used to filter the content are explained in Subsection 6.1. The results rendered by the Map component can be seen in Figure 1.

5.3 Exclusive posts

GeoMedia enables users to interact with the real world by retrieving their current location and then fetching data from the server to display nearby content. When creating a new post (as shown in the left image of Figure 5), a user can choose to make it time-limited by selecting an expiration date and time, or geographically restrict its visibility based on a certain distance, within with other users must be located to view the post. A temporary post will be visible from the moment it is published until the specified expiration time. Additionally, users can define a maximum distance within which the post will be visible, effectively creating a circular area of visibility based on the post's location. Only users within this area —determined by their current GPS coordinates — will be able to see the post. An example of exclusive content is shown in Figure 6. In this case, the user is located in Piazza Bra (Verona, Italy) and is able to view content of various nature, as described in Table 2.

5.4 View post

Another key component is the ViewPost functionality, which displays the content associated with a clicked marker by opening a new interface and fetching the server to retrieve any related multimedia. For efficiency, media files attached to a post are retrieved from the server only when the post is opened. In contrast, the list of visible posts and their corresponding coordinates is requested when the application starts or when the user's current location changes (to detect nearby exclusive posts). This component is not limited to simply viewing the comment; it also allows users to download the attached multimedia files and store them in the local storage of the mobile device. Naturally, the owner of a post can delete the published content using the corresponding button, as shown in Figure 7.

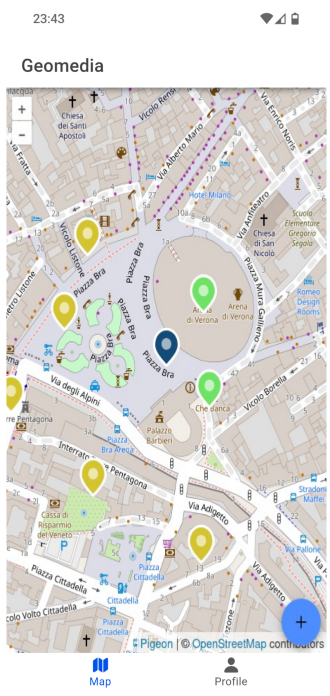

Figure 6: User interface displaying the retrieval of location-based exclusive content. This screen shows how the application detects the user's geographic position to unlock and present content that is only accessible within specific areas. Posts present in the screen are detailed in Table 2.

Number	Type of Post	Type of Content	Exclusive Parameter
5	Multimedia post (yellow marker)	Images	Not applicable
2	Exclusive content (green marker)	Textual	300 meters from Arena di Verona

Table 2: Description of the number of posts, their nature, and visibility parameters corresponding to the content shown in Figure 6.

Figure 7: Screenshot showing the visualization of a post by its creator. Two buttons are available: one to download the attached file (an image of Verona's sunny sky) and another to delete the file.

5.5 HTTP requests

The custom function `doRequest()`, also shown in the snippet of the Map component, sends HTTP requests to the server. Clients communicate with the server using the HTTP protocol, exchanging XMLHttpRequest (XHR) packets that contain data in JSON (JavaScript Object Notation) format, encapsulated in the body of the protocol's `POST` method. To simplify server configuration within the app, the Ionic Storage [18] package has been installed. This package allows data to be stored in the application's cache on the user's mobile device. To enable this feature, it must first be installed via the following terminal command: `npm install @ionic/storage`.

5.6 Ionic Capacitor

To access specific native features of Android OS, such as geolocation or file storage, Ionic Capacitor [17] can be integrated into the Ionic Framework project. Capacitor is a collection of libraries that make possible the use of native components inside web applications, then providing the API to adopt at JavaScript level. GeoMedia requires two specific Capacitor modules, namely:

- geolocation: a plugin used to retrieve the latitude and longitude coordinates of the user currently logged into the app;
- filesystem: a plugin that allows to store files in the mobile file system.

The geolocation package provides the method `getCurrentPosition()` which returns an Object with properties latitude and longitude.

5.7 Building for Android

A core feature of the Ionic Framework is its ability to build web-based applications into native mobile apps: in fact, the framework acts as a wrapper/interpreter for the webpage, essentially embedding a custom browser page inside the app. In order to create an Android Package (APK) for installing the app, few shell commands are required as reported in the official documentation [6]: with the latter commands, the application can be deployed in production mode and then encapsulated into a bundle that will be compiled by Android Studio into the relative SDK.

The Android Manifest is an essential XML file that declares key information about the app to the Android system, including permissions, components, and hardware requirements [3]. In this application, the manifest must be updated to include the location permission, which users must grant upon startup or installation. If denied, the application will not operate correctly. The contents of the Android Manifest are as follows:

- The `<application>` tag includes the following attributes:
  - `android:requestLegacyExternalStorage="true"` – Requests access to legacy external storage APIs.
  - `android:largeHeap="true"` – Requests a larger heap size for memory-intensive applications.
  - `android:usesCleartextTraffic="true"` – Allows cleartext (non-HTTPS) network traffic.
- The application requests the following permissions:
  - `android.permission.ACCESS_COARSE_LOCATION` – For accessing approximate device location.
  - `android.permission.ACCESS_FINE_LOCATION` – For accessing precise device location.
- The application declares the use of the following hardware feature:
  - `android.hardware.location.gps` – Indicates that the app uses GPS hardware.

A possible alternative is using approximate location from other communication systems antennas, although this may significantly reduce accuracy. Moreover, to support cleartext HTTP traffic during development, an explicit exception must be declared in the manifest, as Android 8 and above block non-HTTPS connections by default for security reasons.

6 Backend

The backend consists of an HTTP server, which is the bridge between the stored data and the client application described earlier. The GeoMedia HTTP server uses Node.js as the runtime engine to start a listening process on a designated network port. Furthermore, the server exposes several paths, called endpoints: each of them, when called, triggers the execution of an SQL stored procedure and trigger a backend response. Specifically, input information is passed as parameters, and the resulting data is processed and returned to the client in JSON format. The use of the latter facilitates rapid extensibility of the data structure and provides a format that is easily human-readable. Only the `POST` method is implemented, as every request and response includes data and parameters encapsulated within the body of the HTTP packet. The server relies on a single external dependency that is not included natively: the `tedious` package [34], which can be installed via the command `npm install tedious`. This module is used to establish a connection to a Microsoft SQL Database, to execute query commands and transactions. Finnally another utility, the `Dispatcher` class, is used to wrap the execution of SQL stored procedures and manipulate data when necessary.

6.1 Vincenty's formulae for distance computation

Filtering for exclusive content is handled partially in the application layer. Specifically, the server executes the corresponding SQL stored procedure in the DBMS, which filters out expired posts limited by date. Then, the server calculates the visible radius of each exclusive post and returns only those located within the user's current position. The method used to calculate the distance between the user's location and the position of the limited content—based on a specified radius in kilometers—is an implementation of Vincenty's formulae [1]. For the GeoMedia application, only the first of Vincenty's solutions, known as the direct geodesic problem, has been implemented. This method calculates the geographic coordinates (latitude and longitude) of a destination point given a known starting location, an initial bearing (azimuth), and a geodesic distance. Based on the WGS-84 ellipsoidal model of the Earth, Vincenty's direct formula accounts for the Earth's flattening and provides high-precision results, with an accuracy of up to 0.5 millimeters under normal conditions. This makes it particularly suitable for calculating the visible range of geographically restricted content in the application.

7 Database management system

Following the three-tier design, DBMS represents the Data-layer, providing functionality to store and query data. Microsoft SQL Server (MSSQL) has been selected to implement this role. All posts and user profiles in GeoMedia are stored in dedicated structures, specifically in database tables:

- `USERS` — stores user profiles;
- `POSTS` — stores posts along with all related attributes and metadata.

Data is handled through stored procedures, which provide the data layer with isolation and independence from the rest of the architecture. As a result, future modifications to either the application or data layer are less likely to impact one another. For example, when storing a post, all relevant content and the author's information are passed as parameters, then manipulated and formatted within the appropriate stored procedure inside the DBMS.

8 Future Work

Looking ahead, several promising directions for future development have been identified to enhance both functionality and user experience in GeoMedia. One potential feature is the integration of a Time Capsule system, enabling data to be stored in small physical sensors distributed across urban environments, natural landscapes, or remote regions. These capsules would be accessible exclusively via short-range communication technologies such as Near Field Communication (NFC). This concept supports diverse use cases—for instance, embedding digital waypoints along hiking trails or placing location-bound content at scenic viewpoints, accessible only upon physical presence.

Another enhancement involves the introduction of an altitude-aware sharing system. By utilizing device-generated altitude data, GeoMedia could support vertical content placement and filtering. This would allow for altitude-specific visibility, particularly useful in high-rise buildings, mountainous regions, or drone-assisted applications, thereby adding a novel spatial dimension to media distribution.

In addition to these features, GeoMedia could evolve into a platform for interactive, location-based experiences. Gamified features such as scavenger hunts or augmented hide-and-seek challenges could be integrated, fostering playful exploration and deeper user engagement with geographical spaces. This functionality could be implemented as a default modality within the app, enabling users to instantly participate in or create interactive challenges without additional configuration, thereby encouraging immediate and sustained engagement.

GeoMedia also presents an opportunity for deeper investigation into its role in fostering democratic engagement and civic participation. As a location-based platform that enables grassroots content creation and dissemination, it could empower communities to document local events, express collective concerns, and share context-specific knowledge. In democratic contexts, this may enhance civic transparency and support participatory governance. Conversely, in authoritarian or non-democratic regimes, GeoMedia could serve as a discreet tool for resistance, information sharing, and mobilization, particularly through features like decentralized media storage or physically-bound data access. These possibilities raise important questions around digital freedom, censorship resistance, and ethical safeguards, making this a critical area for future interdisciplinary research and thoughtful platform design.

On the architectural side, a major improvement would involve refactoring the monolithic backend into a microservices-based infrastructure. This transition would enhance scalability, fault tolerance, and maintainability, aligning the system with modern cloud-native development practices. Furthermore, future versions of GeoMedia could incorporate blockchain technology to ensure tamper-proof content verification, decentralized ownership of geotagged media, and enhanced transparency in content provenance. Such integration would be particularly valuable for use cases involving public records, collaborative mapping, or heritage documentation, where content authenticity and traceability are critical.

Finally, a comprehensive testing and iterative development process will be prioritized. This includes usability testing and systematic collection of user feedback, which will be critical in refining the application and ensuring it meets the needs and expectations of a broad user base.

9 Conclusion

GeoMedia provides an open-source, extensible platform for location-based sharing of audio, text, images, and files through an interactive map interface. The envisioned future developments aim to significantly broaden the scope and impact of the platform, incorporating novel features and improving both technical architecture and user experience. With ongoing commitment to innovation and user-centered design, GeoMedia is well-positioned to evolve into a powerful tool for spatial storytelling, community engagement, and geolocated media interaction. More broadly, GeoMedia contributes to the shaping of a digital society in which location-aware technologies foster deeper connections between individuals, communities, and the environments they inhabit. By enabling meaningful, place-based digital interactions, GeoMedia supports the creation of socially enriched, context-sensitive media landscapes that reflect and enhance our collective digital presence in the physical world.

Acknowledgments

This research was carried out within the framework of the project "Sustainable Mobility Center - Centro Nazionale per la Mobilità Sostenibile – CNMS" Spoke 8 "MaaS & Innovative Services" funded by the European Union under the PNRR – MISSIONE 4 Istruzione e Ricerca, COMPONENTE 2: "Dalla ricerca all'impresa", INVESTIMENTO 1.4: "Potenziamento strutture di ricerca e creazione di campioni nazionali di R&S su alcune Key Enabling Technologies", NextGenerationEU, project code CN00000023, thematic area: Sustainable Mobility (Mobilità Sostenibile).

References

[1] T. Vincenty and. 1975. DIRECT AND INVERSE SOLUTIONS OF GEODESICS ON THE ELLIPSOID WITH APPLICATION OF NESTED EQUATIONS. *Survey Review* 23, 176 (1975), 88–93. doi:10.1179/sre.1975.23.176.88 arXiv:https://doi.org/10.1179/sre.1975.23.176.88

[2] Marius Andra. 2025. Pigeon Maps: ReactJS Maps without External Dependencies. https://github.com/mariusandra/pigeon-maps

[3] Android Developers. 2025. App Manifest Overview. https://developer.android.com/guide/topics/manifest/manifest-intro

[4] Shariq Aziz Butt, Mudasser Naseer, Arshad Ali, Abbas Khalid, Tauseef Jamal, and Sumera Naz. 2024. Remote mobile health monitoring frameworks and mobile applications: Taxonomy, open challenges, motivation, and recommendations. *Engineering Applications of Artificial Intelligence* 133 (2024), 108233. doi:10.1016/j.engappai.2024.108233

[5] Armir Bujari, Matteo Ciman, Ombretta Gaggi, and Claudio E Palazzi. 2017. Using gamification to discover cultural heritage locations from geo-tagged photos. *Personal and Ubiquitous Computing* 21 (2017), 235–252.

[6] Capacitor Project. 2025. Capacitor Android Documentation. https://capacitorjs.com/docs/android

[7] Sergio Caro-Alvaro, Eva Garcia-Lopez, Alexander Brun-Guajardo, Antonio Garcia-Cabot, and Aekaterini Mavri. 2024. Gesture-Based Interactions: Integrating Accelerometer and Gyroscope Sensors in the Use of Mobile Apps. *Sensors* 24, 3 (2024), 1004. doi:10.3390/s24031004

[8] Aoxia Chen and Pankaj K. Sen. 2016. Advancement in battery technology: A state-of-the-art review. In *2016 IEEE Industry Applications Society Annual Meeting*. 1–10. doi:10.1109/IAS.2016.7731812

[9] Justin Cranshaw, Raz Schwartz, Jason Hong, and Norman Sadeh. 2021. The Livehoods Project: Utilizing Social Media to Understand the Dynamics of a City. *Proceedings of the International AAAI Conference on Web and Social Media* 6, 1 (Aug. 2021), 58–65. doi:10.1609/icwsm.v6i1.14278

[10] Paula Delgado-Santos, Giuseppe Stragapede, Ruben Tolosana, Richard Guest, Farzin Deravi, and Ruben Vera-Rodriguez. 2022. A Survey of Privacy Vulnerabilities of Mobile Device Sensors. *ACM Comput. Surv.* 54, 11s, Article 224 (Sept. 2022), 30 pages. doi:10.1145/3510579

[11] Fabio Duarte. 2025. Internet Traffic from Mobile Devices (May 2025). https://explodingtopics.com/blog/mobile-internet-traffic

[12] Google. 2025. Google Local Guides. https://maps.google.com/localguides/

[13] Matthew Halpern, Yuhao Zhu, and Vijay Janapa Reddi. 2016. Mobile CPU's rise to power: Quantifying the impact of generational mobile CPU design trends on performance, energy, and user satisfaction. In *2016 IEEE International Symposium on High Performance Computer Architecture (HPCA)*. 64–76. doi:10.1109/HPCA.2016.7446054

[14] Zoltan Horvath, Ildiko Jenak, and Ferenc Brachmann. 2017. Battery consumption of smartphone sensors. *Journal of Reliable Intelligent Environments* 3, 2 (01 Aug 2017), 131–136. doi:10.1007/s40860-017-0034-1

[15] International Telecommunication Union. 2023. Facts and Figures 2023 – Mobile Phone Ownership. https://www.itu.int/itu-d/reports/statistics/2023/10/10/ff23-mobile-phone-ownership/

[16] Ionic Framework. 2025. Ionic Framework: The Cross-Platform App Development Leader. https://ionicframework.com/

[17] Ionic Team. 2025. Capacitor: Build Cross-Platform Native Progressive Web Apps. https://github.com/ionic-team/capacitor

[18] Ionic Team. 2025. Ionic Storage: A Simple key-Value Storage Module for Ionic Apps. https://github.com/ionic-team/ionic-storage Accessed: May 31, 2025.

[19] Seung Ryul Jeong and Hye Jin Yoon. 2015. A study on Factors that Influence the Usage of Mobile Apps – Based on Flow Theory and Unified Theory of Acceptance and Use of Technology. *International Journal of Multimedia and Ubiquitous Engineering* 10, 6 (2015), 93–104. doi:10.14257/ijmue.2015.10.6.10

[20] Naveen Kumar. 2025. Facebook Users Statistics (2025) — New Worldwide Data. https://www.demandsage.com/facebook-statistics/

[21] Oksana Kyrylova, Neliia Blynova, and Viktoriia Pavlenko and. 2024. The perspectives for mobile application use in media education. 3592–3599 pages. doi:10.1080/10494820.2023.2186897 arXiv:https://doi.org/10.1080/10494820.2023.2186897

[22] Allan Martin. 2008. Digital literacy and the "digital society". *Digital literacies: Concepts, policies and practices* 30, 151 (2008), 1029–1055.

[23] Raluca Ionela Maxim and Joan Arnedo-Moreno. 2025. Identifying key Principles and Commonalities in Digital Serious Game Design Frameworks: Scoping Review. *JMIR Serious Games* 13 (5 Mar 2025), e54075. doi:10.2196/54075

[24] Alberto Morini. 2025. GeoMedia: A Geolocation Android App (and Server) for UNIPD. https://github.com/albertomorini/GeoMedia

[25] Shree Om and William D. Tucker. 2018. Battery and Data Drain of Over-The-Top Applications on Low-end Smartphones. In *2018 IST-Africa Week Conference (IST-Africa)*. Page 1 of 13–Page 13 of 13.

[26] Catia Prandi, Marco Roccetti, Paola Salomoni, Valentina Nisi, and Nuno Jardim Nunes. 2017. Fighting exclusion: a multimedia mobile app with zombies and maps as a medium for civic engagement and design. *Multimedia Tools and Applications* 76, 4 (01 Feb 2017), 4951–4979. doi:10.1007/s11042-016-3780-9

[27] Alexandru Predescu and Mariana Mocanu. 2024. Gamification in Real-World Applications: Interactive Maps and Augmented Reality. In *Level Up! Exploring Gamification;s Impact on Research and Innovation*, Tibor Guzsvinecz (Ed.). IntechOpen, Rijeka, Chapter 2. doi:10.5772/intechopen.1004870

[28] Oliver Roick and Susanne Heuser. 2013. Location Based Social Networks – Definition, Current State of the Art and Research Agenda. *Transactions in GIS* 17 (05 2013). doi:10.1111/tgis.12032

[29] Katarzyna Siła-Nowicka, Jan Vandrol, Taylor Oshan, Jed A. Long, Urška Demšar, and A. Stewart Fotheringham and. 2016. Analysis of human mobility patterns from GPS trajectories and contextual information. *International Journal of Geographical Information Science* 30, 5 (2016), 881–906. doi:10.1080/13658816.2015.1100731 arXiv:https://doi.org/10.1080/13658816.2015.1100731

[30] Statista. 2014. 30% of Facebook Users Are Now Mobile-Only. https://www.statista.com/chart/2494/facebooks-users-by-type-of-access/

[31] Statista. 2023. Global smartphone sales to end users from 2007 to 2023. https://www.statista.com/statistics/263437/global-smartphone-sales-to-end-users-since-2007/

[32] Statista. 2024. Most popular social networks worldwide as of January 2024, ranked by number of monthly active users. https://www.statista.com/statistics/272014/global-social-networks-ranked-by-number-of-users/

[33] Marcin Straczkiewicz, Peter James, and Jukka-Pekka Onnela. 2021. A systematic review of smartphone-based human activity recognition for health research. *npj Digital Medicine* 4, 1 (2021), 1–16. doi:10.1038/s41746-021-00514-4

[34] Tedious Project. 2025. Tedious: Node.js TDS Driver for Microsoft SQL Server. https://tediousjs.github.io/tedious/

[35] Leonardo Tonetto, Pauline Kister, Nitinder Mohan, and Jörg Ott. 2024. Ethical and Privacy Considerations with Location Based Data Research. arXiv:2403.05558 [cs.CY] https://arxiv.org/abs/2403.05558

[36] W3Schools. 2025. SQL Stored Procedures. https://www.w3schools.com/sql/sql_stored_procedures.asp Accessed: May 31, 2025.

[37] Chao Zhang, Lidan Shou, Ke Chen, Gang Chen, and Yijun Bei. 2012. Evaluating geo-social influence in location-based social networks. In *Proceedings of the 21st ACM International Conference on Information and Knowledge Management* (Maui, Hawaii, USA) *(CIKM '12)*. Association for Computing Machinery, New York, NY, USA, 1442–1451. doi:10.1145/2396761.2398450

[38] Yu Zheng and Xing Xie. 2011. *Location-Based Social Networks: Locations.* Springer New York, New York, NY, 277–308. doi:10.1007/978-1-4614-1629-6_9